SIAM AREFIN

Email | GitHub | LinkedIn | Portfolio

Research Interests: Machine Learning, Computer Vision, Large Language Models, Generative AI

Academic Credentials

Shahjalal University of Science and Technology (SUST), Bangladesh Mar 2022 – Jul 2026
Bachelor of Science in Software Engineering ***CGPA: 3.10/4.00***

Research Experiences

LLDDet – An Annotated Lemon Leaf Disease Detection Dataset using Deep Learning

- Developed a deep learning-based object detection system for identifying and localizing diseases in lemon leaves using the LLDDet dataset.
- Achieved a test mAP of **0.927** while training under constrained computational resources with a batch size of 4 for 27 epochs.
- Investigated model generalization and object detection performance for real-world agricultural disease identification.

omicML: An Integrative Bioinformatics and Machine Learning Framework for Transcriptomic Biomarker Identification *Machine Learning Research Engineer | Collaboration with Department of Biochemistry and Molecular Biology (BMB), SUST*

- Implemented scalable machine learning deployment and monitoring pipelines using MLOps best practices.
- Collaborated with bioinformatics researchers to integrate computational modeling with biological large-data analysis.
- Worked on an interdisciplinary research problem connecting machine learning with transcriptomic biomarker identification.

Publications

- **Siam Arefin**, "LLDDet: Lemon Leaf Disease Detection using Deep Learning," *Manuscript in Preparation*, 2026.
- Joy Prokash Debnath, Kabir Hossen, Md. Sayeam Khandaker, Shawon Majid, Md. Mehrajul Islam, **Siam Arefin**, Preonath Chondrow Dev, Saifuddin Sarker, Tanvir Hossain, "omicML: An Integrative Bioinformatics and Machine Learning Framework for Transcriptomic Biomarker Identification," *bioRxiv*, 2025.

  doi:10.1101/2025.10.25.684517

Industry Experience

Kite Games Studio Nov 2025 – Jan 2026
Software Engineer (ML Team) Intern | Mohakhali DOHS, Dhaka — Onsite

- Conducted R&D in image processing, machine learning, and generative AI for production-oriented applications.
- Worked with Diffusion and FLUX-based generative AI models for image generation and processing.
- Contributed to an Android application with 5M+ downloads and 700K+ daily active users.

Academic and Competitive Achievements

- **1st Place – Multi-Instance Object Detection Challenge**, Duality AI International Competition; developed YOLO-based detection with ensemble learning and Sim2Real adaptation, achieving **0.995 private leaderboard score**.
- **1st Runner-up – ML Olympiad**, CO_2 Emissions Prediction Challenge; developed an ML pipeline involving preprocessing, feature engineering, model training, and optimization.
- **Top 10 – DL Sprint 3.0**, Bengali AI Math Olympiad; developed Prompt Engineering-based solutions for Bengali mathematical problem with Qwen model.
- **4th Place – DL Enigma 1.0**, SUST CSE Carnival 2024; developed and optimized a YOLOv11m model for vehicle detection task.
- **800+ Problems Solved** on Codeforces and other competitive programming platforms, covering algorithms, data structures, and problem solving.

Technical Skills

Machine Learning:	Machine Learning, Deep Learning, NLP, CNNs, RNNs, Transformers, PyTorch, TensorFlow, Feature Engineering
Generative AI / LLMs:	LLMs, Prompt Engineering, Fine-tuning, RAG, Vector Databases, LangChain, AI Agents, Hugging Face, Diffusion Models
Computer Vision:	OpenCV, YOLO, Object Detection, Image Processing and Generation, Ensemble Learning
Programming:	Python, C, C++, Java, JavaScript
AI Systems / MLOps:	FastAPI, REST APIs, Docker, Git/GitHub, CI/CD, Azure, SQL, MongoDB
Software Engineering:	OOP, SOLID, Design Patterns, DSA, Software Architecture, Unit Testing